Optimizing Multi-Class Chest X-Ray Pathology Detection under Asymmetric Clinical Costs

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***Abstract*—Automated diagnosis of thoracic diseases from chest X-rays is challenging due to subtle visual patterns and severe class imbalance. This work addresses a 20-class pathology classification task with an asymmetric cost function that penalizes False Negatives five times more than False Positives.**

We propose a cost-aware deep learning pipeline based on DenseNet201. Progressive resizing accelerates training, while a weighted sampler mitigates the dominance of the "No Finding" class. Post-hoc threshold optimization aligns predictions with the asymmetric metric, and ensembling stratified cross-validation models preserves calibrated probability distributions.

Our inference pipeline combines Test-Time Augmentation, temperature scaling, and a margin rule. The approach achieved a score of -4.28660 on the competition public leaderboard (65% of the test set), demonstrating the effectiveness of cost-aware calibration for medical image classification.

***Index Terms*—Chest X-ray, DenseNet201, Progressive Resizing, Asymmetric Cost, Precision-Recall Trade-off, Calibration.**

I. Introduction

A. Clinical Motivation and Problem Statement

Chest radiography (CXR) is a widely used diagnostic tool for thoracic diseases, but manual interpretation is labor-intensive and prone to variability [1]. Deep learning–based Computer-Aided Diagnosis systems can assist radiologists by automatically detecting critical pathologies [5].

This work addresses a 20-class pathology classification task with two challenges:

1) **Extreme Class Imbalance:** The majority class, "No Finding" (Class 19), represents 66% of training data (≈45k images), while several classes contain fewer than 100 samples.
2) **Asymmetric Cost Optimization:** False Negatives are far more harmful than False Positives. The evaluation metric reflects this asymmetry: True Positive (+1), False Positive (-1), False Negative (-5).

Accuracy-driven models tend to predict "No Finding", reducing FP penalties but producing excessive FNs. We therefore design a pipeline that directly optimizes the asymmetric metric via precision–recall trade-off and cost-aware thresholding.

II. Methodology

A. Dataset and Preprocessing

The dataset contains $N_{train} = 68{,}058$ training images and $N_{test} = 20{,}279$ test images.

1) Normalization: Chest X-ray images are single-channel (grayscale). To use ImageNet-pretrained architectures, the channel is replicated to form 3-channel tensors $X \in \mathbb{R}^{3\times H\times W}$. Each channel is normalized using ImageNet statistics:

$$X_{norm}^{(c,i,j)} = \frac{X^{(c,i,j)} - \mu_c}{\sigma_c} \tag{1}$$

where $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$.

2) Stochastic Data Augmentation: To improve generalization, the input tensor undergoes stochastic augmentations:

$$\tilde{X} = \mathcal{T}_{color}\big(\mathcal{T}_{affine}(\mathcal{T}_{flip}(X))\big) \tag{2}$$

The affine transformation is defined as

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{3}$$

where $\theta \sim \mathcal{U}(-10^\circ, 10^\circ)$ and translations $t_x, t_y \sim \mathcal{U}(-0.05, 0.05)$ of image dimensions [3]. Random horizontal flips ($p = 0.5$) and mild color jitter are also applied.

B. Model Architecture: DenseNet201

We adopt **DenseNet201** [4] as the backbone. Unlike feed-forward architectures such as ResNet, DenseNet connects each layer to all preceding layers. For layer l:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}]) \tag{4}$$

where $[\cdot]$ denotes channel-wise concatenation and $H_l(\cdot)$ consists of Batch Normalization, ReLU, and a 3×3 convolution.

Dense connections provide two advantages:

1) **Improved Gradient Flow:** Direct connections facilitate gradient propagation and mitigate vanishing gradients.
2) **Feature Reuse:** Low-level texture features important for detecting subtle thoracic abnormalities are preserved across network depth.

The global average pooling layer outputs feature vectors $f \in \mathbb{R}^{1920}$. To reduce overfitting, Dropout is applied before the classification layer:

$$z = W(\text{Dropout}(f, p = 0.4)) + b \tag{5}$$

where $W \in \mathbb{R}^{20\times 1920}$ and z are the output logits.

C. Training Strategy: Progressive Resizing

Training CNNs on high-resolution images is computationally expensive. We employ **Progressive Resizing** [2], a curriculum learning strategy where image resolution is gradually increased during training:

- **Stage 1** (224×224 **px):** 3 epochs. The backbone is frozen for the first epoch to stabilize the classification head.
- **Stage 2** (320×320 **px):** 2 epochs of full fine-tuning.
- **Stage 3** (384×384 **px):** 2 epochs of final refinement.

This enables learning coarse structural features (e.g., heart size, lung boundaries) at lower resolution before refining subtle textures (e.g., localized nodules) at higher resolution.

D. Addressing Imbalance: Weighted Random Sampler

To mitigate extreme class imbalance, we use a **Weighted Random Sampler**. Let N_{total} be dataset size and N_c frequency. Sampling weights are inversely proportional:

$$W_c = \min\left(10, \max\left(0.5, \frac{N_{total}}{20 \cdot N_c}\right)\right) \tag{6}$$

Because "No Finding" class dominates dataset, its weight is reduced ($W_{19} \leftarrow 0.5\, W_{19}$) to limit influence on gradients.

E. Objective Function: Label Smoothed Cross-Entropy

Standard Cross-Entropy drives logits $z_i \to \infty$ for the correct class, producing overconfident predictions. We apply **Label Smoothing** ($\epsilon = 0.1$) [10] to one-hot targets y:

$$y_i^{ls} = (1 - \epsilon) y_i + \frac{\epsilon}{20} \tag{7}$$

We optimize using AdamW [8] ($lr = 3e{-}4$, weight decay $1e{-}4$) with Cosine Annealing [6]:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{20} y_i^{ls} \log\left(\frac{e^{z_i}}{\sum e^{z_j}}\right) \tag{8}$$

F. Theoretical Bayes Risk & Threshold Optimization

Metric is macro-averaged asymmetric score. For class c:

$$\text{Score}_c = \frac{TP_c - FP_c - 5\, FN_c}{N_{true,c}} \tag{9}$$

Standard Softmax classification uses `argmax`, implying a decision boundary at $p = 0.5$. In cost-sensitive learning [9], the Bayes boundary minimizes expected risk. Let $C_{FP} = 1$ and $C_{FN} = 5$. A positive prediction occurs when:

$$p\, C_{FP} < (1 - p)\, C_{FN} \implies p > \frac{C_{FP}}{C_{FP} + C_{FN}} \tag{10}$$

Substituting the penalties yields the theoretical threshold:

$$t_{theory} = \frac{1}{1 + 5} \approx 0.166 \tag{11}$$

Since probabilities deviate from theory, we perform **Greedy Threshold Search**. Using out-of-fold P, we select $t_c \in [0.01, 0.99]$ maximizing the metric:

$$t_{opt}^c = \operatorname*{argmax}_t \left(\frac{TP_c(t) - FP_c(t) - 5\, FN_c(t)}{N_{true,c}}\right) \tag{12}$$

G. Inference and Ensembling Strategy

Post-hoc threshold optimization depends on $P_{CV}(X)$ from cross-validation. Retraining on the full dataset shifts outputs to $P_{full}(X)$, causing calibration mismatch with thresholds tuned on P_{CV}. Final logits are averaged across the four CV models:

$$L_{ensemble} = \frac{1}{4} \sum_{i=0}^{3} L_{Fold\_i} \tag{13}$$

We applied **Test-Time Augmentation (TTA)** across three views (Original, Horizontal Flip, Rotation $+10°$) to reduce aleatoric uncertainty.

Finally, to counter over-peaked distributions & allow smooth thresholding, we do **Temperature Scaling** ($T_{scale} = 1.4$) [7]:

$$p_i = \frac{\exp(L_i / T_{scale})}{\sum_j \exp(L_j / T_{scale})} \tag{14}$$

We introduce a **Margin Rule**: if the probability gap between the top-1 and top-2 predictions is small ($p_{top1} - p_{top2} < 0.08$), we override optimized thresholds and use `argmax` to avoid false positives in highly ambiguous test cases.

III. Experimentation

We use **Stratified 4-Fold Cross-Validation**, ensuring rare classes are evenly distributed across validation sets.

Training Dynamics: Figure 1 shows training and validation accuracy across folds. The step changes at Epochs 3 and 5 correspond to progressive resizing transitions. Although accuracy briefly dips during resolution changes, it later converges to a higher plateau, validating the multi-scale training strategy.

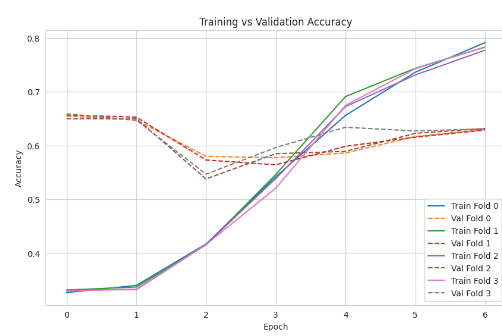

Fig. 1. Training vs. Validation Accuracy. The stepwise improvement mathematically validates the Progressive Resizing strategy across learning stages.

IV. Results and Analysis

Table I summarizes cross-validation performance. The low inter-fold variance ($\sigma_{AUC} < 0.02$) indicates minimal dataset-specific overfitting.

A. Diagnostic Performance (Per-Class AUC-ROC)

We analyze ROC (Figure 2) and AUC (Figure 3) curves to measure class-wise discrimination independent of thresholds.

- **High-Separability: Class 4 (Cardiomegaly)** achieved the highest AUC (≈ 0.89), likely because the enlarged heart silhouette forms a clear macro-level spatial feature.

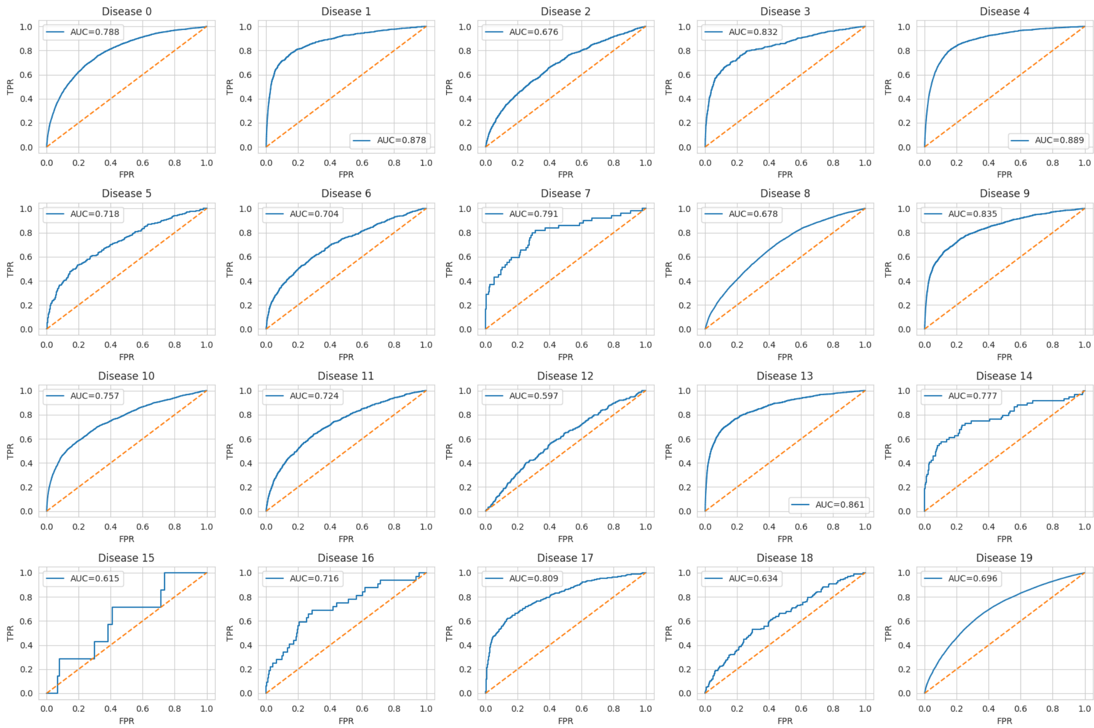

Fig. 2. **Per-Class AUC-ROC**: For all 20 pathologies. Cardiomegaly (Class 4) shows high separability, while Class 12 approaches near-linear behavior.

TABLE I
CROSS-VALIDATION RESULTS (DENSENET201)

Fold	Best Asym Score	**Macro F1**	**ROC-AUC**	**Recall**
0	-4.430	0.186	0.735	0.192
1	-4.659	0.168	0.770	0.183
2	-4.537	0.172	0.741	0.175
3	-4.463	0.174	0.754	0.176
Mean	**-4.522**	**0.175**	**0.750**	**0.181**

- **Low-Separability: Class 12** was most challenging (AUC $\approx$ 0.59), likely reflecting subtle pathologies (e.g., small nodules) with weak signals masked by bone textures.

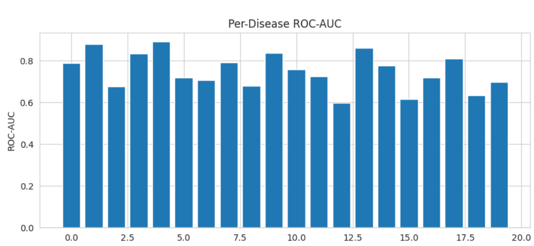

Fig. 3. **Per-Class AUC-ROC** bar chart showing variance across classes.

B. Threshold Analysis: The Precision-Recall Trade-off

Recall (Sensitivity) is critical in medical applications as it reflects the False Negative rate. As shown in Table I, baseline Macro Recall is about 18.1% before threshold optimization. Our **Threshold Analysis** (Figure 4) optimizes the **precision–recall trade-off** under the asymmetric metric.

The optimal threshold set $T = \{t_1, \ldots, t_{20}\}$ is strongly **bimodal**, illustrating this trade-off:

1) **Aggressive Hyperplanes ($t < 0.1$):** Due to heavy FN penalty (-5), model sacrifices precision & flags disease even if $P(Y = 1|X) = 0.01$, massively boosting **Recall**.
2) **Conservative Hyperplanes ($t > 0.8$):** For low-AUC classes (e.g., 12), predictions yield more FP (-1) than TP (+1), so thresholds remain high to favor **Precision**.

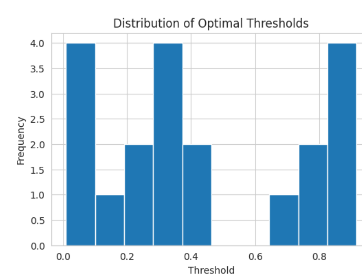

Fig. 4. **Threshold analysis** showing optimal threshold distribution. The bimodal pattern indicates a uniform 0.5 threshold fails under asymmetric costs.

Figure 5 shows the precision–recall trade-off as thresholds vary across classes. Lower thresholds increase recall but reduce precision due to more false positives. Dashed lines mark thresholds selected via cross-validation. Well-separated classes (e.g., Cardiomegaly) show smooth curves, while rare diseases exhibit unstable behavior from severe imbalance.

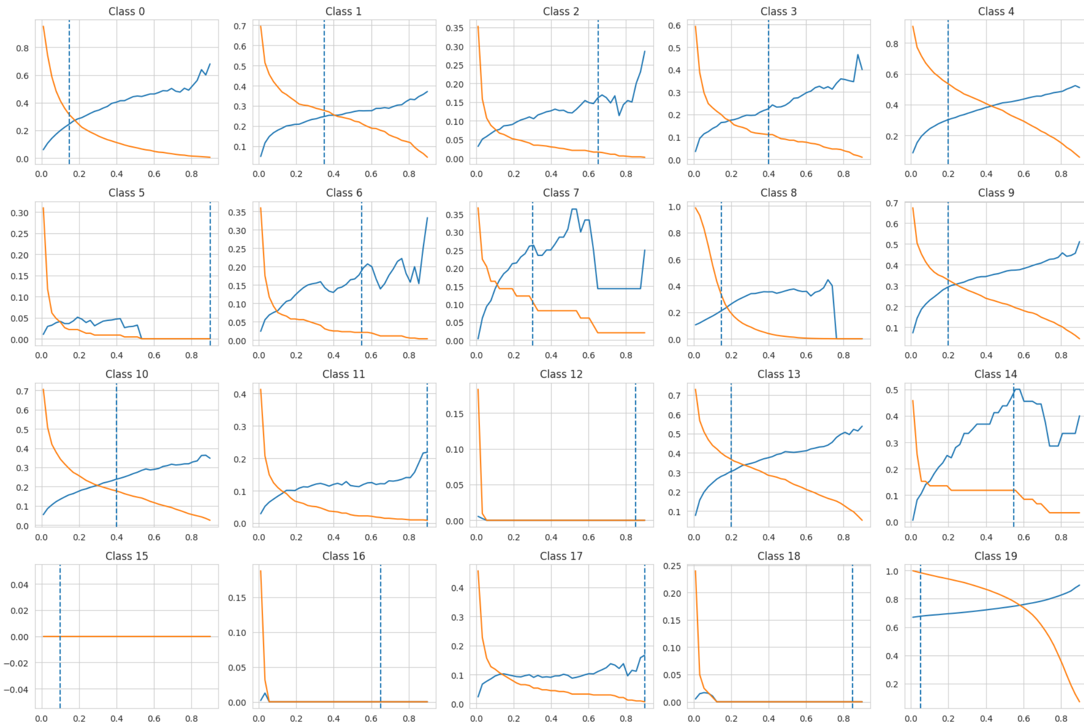

Fig. 5. **Per-Class Precision–Recall Trade-off**. Precision increases and recall decreases as thresholds rise. Dashed lines mark optimized thresholds from cross-validation. Well-separated classes show smooth curves, while rare classes exhibit unstable behavior due to severe imbalance.

Calibration curves (Figure 6) lie below the $y = x$ line, indicating under-confident predictions [7] and motivating lower post-hoc decision thresholds.

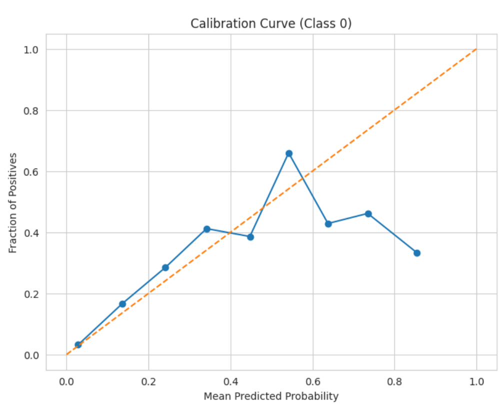

Fig. 6. Calibration Curve for Class 0. The deviation below the diagonal mathematically validates the shift toward lower decision boundaries.

C. Final Competition Performance

Our 4-fold CV ensemble with Test-Time Augmentation, Temperature Scaling, and optimized thresholds achieved a score of **-4.28660** on the Public Leaderboard (65% of the hidden test set).

V. Conclusion

We presented a cost-aware deep learning pipeline for multi-class chest X-ray pathology detection under asymmetric clinical costs. By combining DenseNet201, progressive resizing, calibration-aware CV ensembling, and per-class threshold optimization based on Bayes risk, we effectively controlled the precision–recall trade-off and aligned predictions with clinical penalties. Future work includes exploring hierarchical losses for modeling relationships between thoracic diseases.

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